Zomato Restaurant Data Analysis

UST — 100 Next Gen AI Engineer Cohort 3 | Assignment 1 • Exploratory Data Analysis (Pandas, NumPy, Matplotlib, Seaborn)

1. Dataset Overview

The dataset lists restaurants on Zomato with details on location, cuisines, pricing, ratings and customer engagement. It spans multiple countries but is dominated by India. The raw file contained **9,552 rows × 21 columns**. Key columns include Restaurant Name, City, Cuisines, Average Cost for two, Has Online delivery, Has Table booking, Price range, Aggregate rating, Rating text and Votes.

2. Data Cleaning

• **Malformed records:** 2 rows were corrupted — a Singapore restaurant ("Super Loco") whose address contained embedded commas/line-breaks broke CSV parsing and shifted values across columns. These were dropped, leaving **9,550 rows × 21 columns**.

• **Missing values:** after cleaning, only **Cuisines** has missing entries (9 rows).

• **Data type changes:** `Average Cost for two`, `Votes` and `Price range` were stored as text (object) and converted to numeric types.

• **Duplicates:** none — 0 exact duplicate rows and 0 duplicate Restaurant IDs.

• **Sentinel values:** an Aggregate rating of **0** means "Not rated" (2,148 restaurants). Rating averages are reported both overall and for rated-only restaurants.

3. Data Analysis — Statistical Summary & Answers

Question	Answer
Unique cities	141
City with most restaurants	New Delhi (5,473)
Most frequent cuisine	North Indian (3,960 listings)
Average rating (rated only / all rows)	3.44 / 2.67
Most-voted restaurant	Toit, Bangalore (10,934 votes)
Average cost for two	1,199.33 (mixed currency, mostly INR)
Avg rating: Online delivery Yes / No (rated)	3.381 / 3.467
Avg rating: Table booking Yes / No (rated)	3.588 / 3.414

Top 10 cities by restaurant count: New Delhi (5,473), Gurgaon (1,118), Noida (1,080), Faridabad (251), Ghaziabad (25), Bhubaneshwar (21), Lucknow (21), Ahmedabad (21), Amritsar (21), Guwahati (21).

Key observation: the market is highly concentrated in Delhi NCR (New Delhi, Gurgaon, Noida, Faridabad), which together account for the large majority of records.

4. Data Visualizations

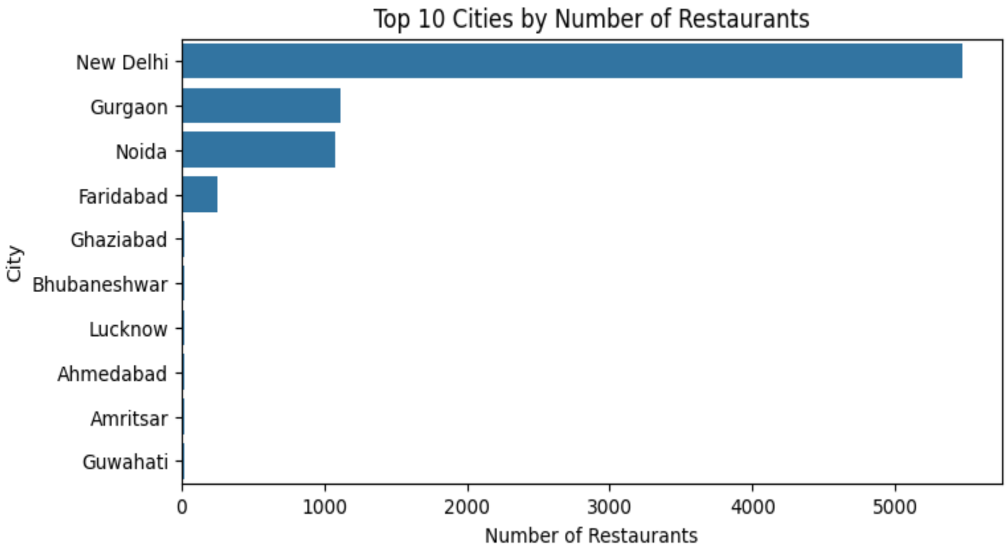

Fig 1. Delhi NCR dominates: New Delhi alone has 5,473 restaurants, with Gurgaon and Noida next.

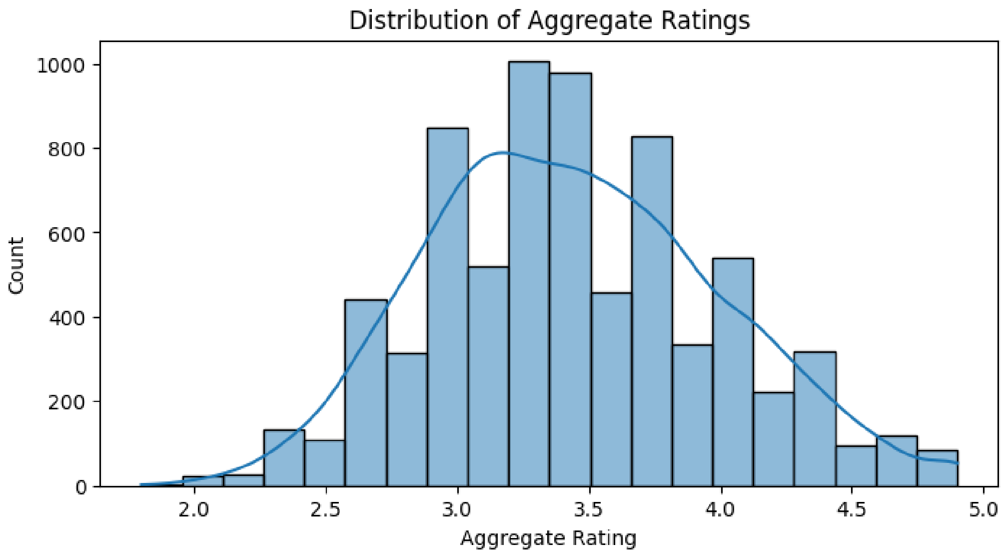

Fig 2. Ratings of rated restaurants are roughly bell-shaped, centered around 3.2-3.7; few extremes.

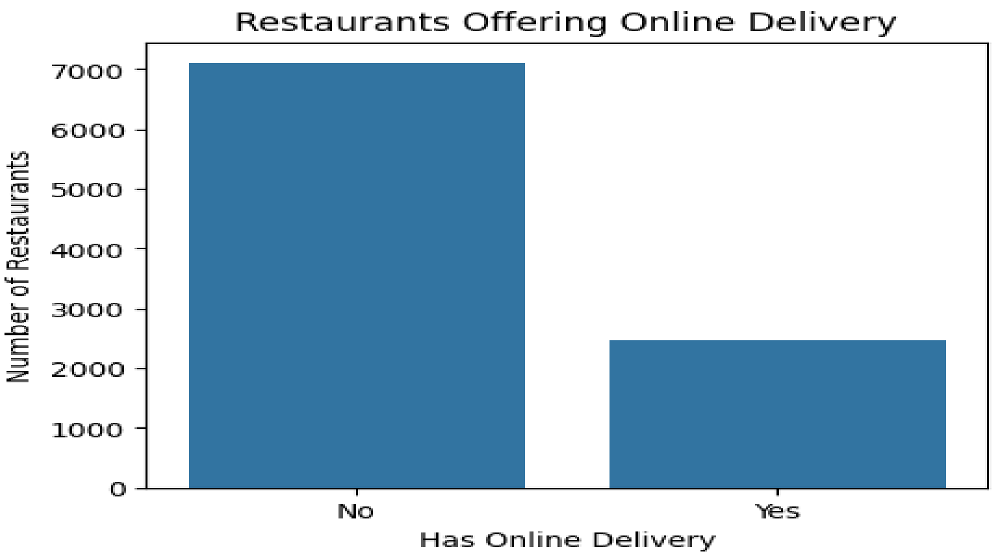

Fig 3. Only ~26% of restaurants (2,451) offer online delivery; most are dine-in / non-delivery.

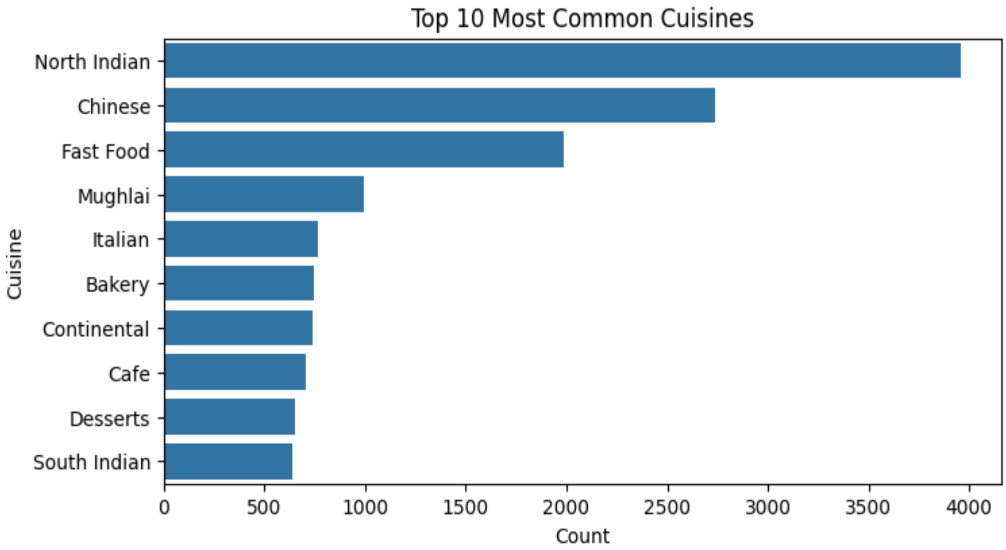

Fig 4. North Indian and Chinese lead by a wide margin, reflecting the India-heavy dataset.

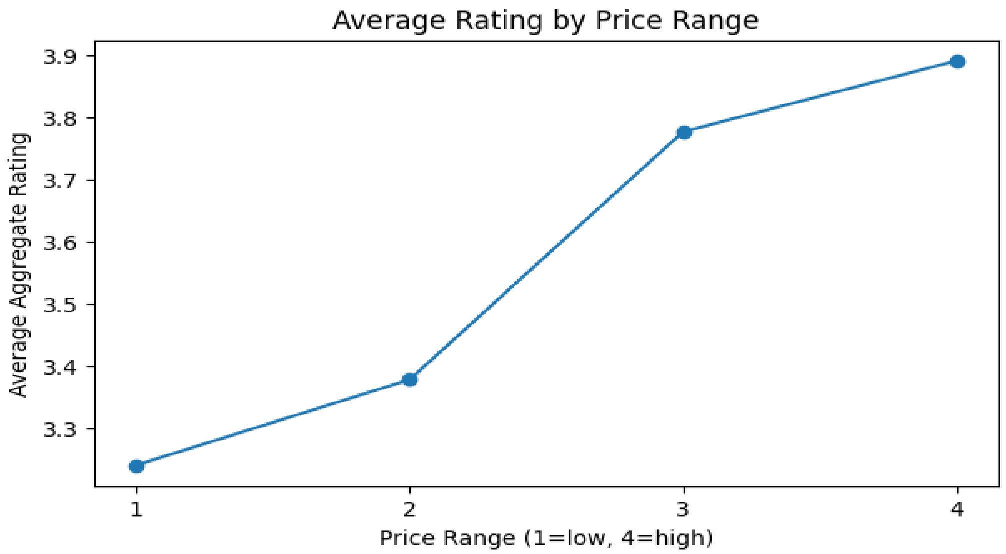

Fig 5. Average rating rises steadily with price range, from ~3.24 (range 1) to ~3.89 (range 4).

5. Conclusion — Insights & Business Observations

1. **Highest restaurant presence:** New Delhi (5,473 restaurants); the Delhi NCR cluster dominates the data.

2. **Most popular cuisine:** North Indian, followed by Chinese and Fast Food.

3. **Online delivery vs ratings:** No genuine advantage. The higher raw average for delivery restaurants (3.249 vs 2.465) is a composition effect — most unrated (0) restaurants are non-delivery. Among rated restaurants the averages converge (3.381 vs 3.467).

4. **Table booking vs ratings:** Yes — restaurants offering table booking rate slightly higher (3.588 vs 3.414 among rated), likely because they skew toward higher-end establishments.

5. **What I learned:** Real-world data demands careful cleaning (malformed rows, wrong dtypes, and "0" sentinel values for unrated restaurants). Naive averages can mislead, so segmenting rated vs unrated matters. The dataset is geographically skewed toward India, and rating correlates positively with price range.